

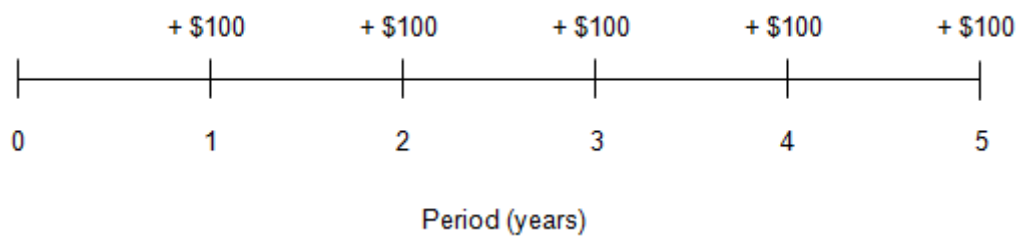
Time Value of Money

The time value of money (TVM) is the concept that a sum of money is worth more now than the same sum will be at a future date due to its earnings potential in the interim. The time value of money is a core principle of finance. A sum of money in the hand has greater value than the same sum to be paid in the future. The time value of money is also referred to as the present discounted value.

Time Lines:

A series of cash flows can be graphically represented using a *cash flow timeline*. A timeline depicts the timing and amount of the cash flows.

For example, the following timeline depicts cash inflows of \$100 to be received at the end of each of the next 5 years:



Cash flows in a timeline are often labeled positive or negative. By convention, positive cash flows correspond to cash inflows; negative cash flows correspond to cash outflows.

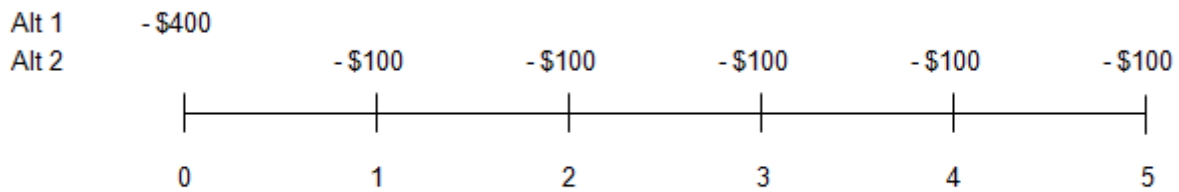
Whether a cash flow is an inflow or an outflow depends on perspective (i.e., as a borrower or lender). The borrower's inflow is the lender's outflow, and vice versa.

In using timelines, and in solving time value of money problems, one should adopt the perspective of either borrower or lender and stay with that perspective throughout the problem.

Consider a simple time value of money problem. In making a purchase you are given two payment alternatives:

- Pay \$400 immediately.
- Pay five installments of \$100 each at the end of each of the next five years.

As depicted on a cash flow timeline:



In deciding which alternative is better, we can't simply add up the five payments of \$100 and compare this sum (\$500) with alternative 1 (\$400 today). To do so would ignore the time value of money because the two alternatives involve payments at different times. Instead, we must determine the value today (at time 0) of the five future payments of \$100 of alternative 2 and compare this to \$400, which is the value today of alternative 1. As we shall see, determining the value today (the present value) of the five payments under alternative 2 involves calculating the present value of those payments at a given rate of interest.

Time value of money calculations allow us to solve problems such as the one above and many others.

Future Values:

The $FW\$1$ is the amount to which \$1 will grow at periodic interest rate i after n periods, assuming the payment of \$1 occurs at the beginning of the first period.

The $FW\$1$ is used to compound a single present amount to its future amount. The future worth of 1 factor ($FW\$1$) is based on the premise that \$1 deposited at the beginning of a period earns interest during the period and becomes part of the principal at the beginning of the next period. This continues for the number of periods in the problem.

Future Value Calculations:

The formula for the calculation of the $FW\$1$ factors is

$$FW\$1 = (1 + i)^n$$

Where:

- $FW\$1$ = Future Worth of \$1 Factor
- i = Periodic Interest Rate, often expressed as an annual percentage rate

- n = Number of Periods, often expressed in years

All of the other compound interest formulas published in AH 505 are derived from the basic compounding expression in the $FW\$1$ factor, $(1 + i)^n$. As we will see, this mathematical expression is the basic building block of all the other compound interest formulas.

The periodic interest rate, i , must match the compounding period, n (this holds for all compound interest functions).

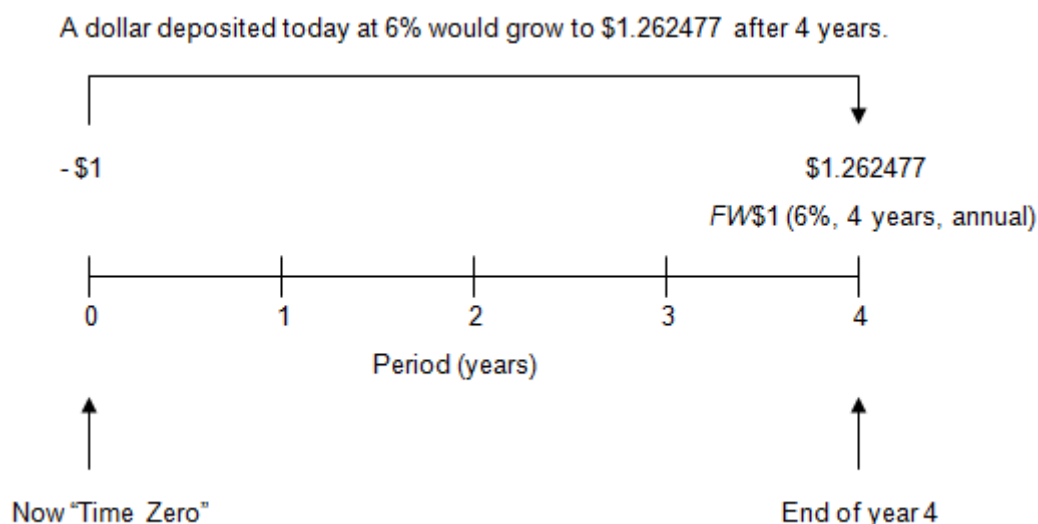
For example, if n is stated in years, indicating annual compounding, i must be stated as an annual rate; if n is stated in months, indicating monthly compounding, i must be stated as a monthly rate.

For now, we will assume annual compounding, so our periods, n , will be in years and the periodic interest rate, i , will be the annual percentage rate. Later, we will introduce the concept of more than one compounding period per year (monthly, quarterly, etc.).

In order to calculate the annual $FW\$1$ factor for 4 years at an annual interest rate of 6%, use the formula below:

- $FW\$1 = (1 + i)^n$
- $FW\$1 = (1 + 0.06)^4$
- $FW\$1 = (1.06)^4$
- $FW\$1 = 1.262477$

Viewed on a timeline:



On the timeline, the initial deposit of \$1 is shown as negative because from the point of view of a depositor it would be a cash outflow. The future value is shown as positive because it would be a cash inflow. The depositor gives up money now in order to receive money later.

Present Values:

The $PW\$1$ factor is the amount that must be deposited today to grow to \$1 in the future, given periodic interest rate i and n periods.

The $PW\$1$ factor is used to discount a single future amount to its present amount. The $PW\$1$ factor can be thought of as the opposite of the $FW\$1$ factor; mathematically, the $PW\$1$ and $FW\$1$ factors are reciprocals:

$$PW\$1 = \frac{1}{FW\$1}$$

Whereas the $FW\$1$, discussed in Lesson 1, provides the future value of a single present amount, the $PW\$1$ provides the present value of a single future amount.

The value of the $PW\$1$ factor will always be less than \$1, explicitly demonstrating that a dollar to be received in the future is worth less than a dollar today.

Present Value Calculations:

The formula for the calculation of the $PW\$1$ factors is:

$$PW\$1 = \frac{1}{(1 + i)^n}$$

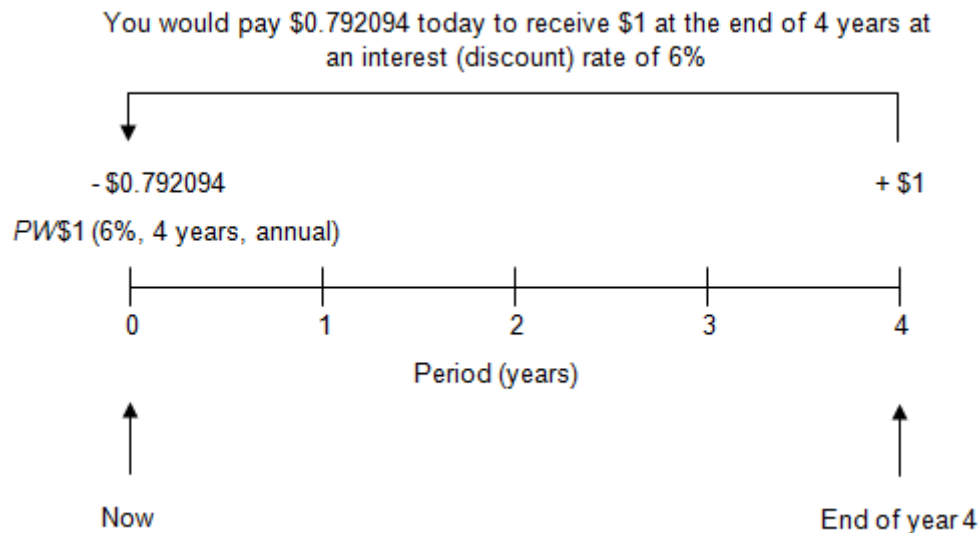
Where:

- $PW\$1$ = Present Worth of \$1 Factor
- i = Periodic Interest Rate, often expressed as an annual percentage rate
- n = Number of Periods, often expressed in years

In order to calculate the $PW\$1$ factor for 4 years at an annual interest rate of 6%, use the formula below:

$$\begin{aligned}
 PW\$1 &= \frac{1}{FW\$1} \\
 &= \frac{1}{(1+i)^n} \\
 &= \frac{1}{(1+0.06)^4} \\
 &= \frac{1}{1.262477} \\
 &= 0.792094
 \end{aligned}$$

Viewed on a timeline:



A depositor would be willing to give up \$0.792094 today (shown as negative on the timeline) in order to receive \$1 at the end of 4 years (shown as positive).

Find the Interest Rate, I

To find Rate when Principal Interest and Time are given the rules are.

$$\text{Interest} = (\text{Principal} \times \text{Rate} \times \text{Time})/100$$

$$\text{Rate} = (100 \times \text{Interest})/(\text{Principal} \times \text{Time})$$

Examples to find Rate when Principal, Interest and Time are given:

1. Find Rate, when Principal = \$ 3000; Interest = \$ 400; Time = 3 years.

Solution:

Principal = \$ 3000,

Interest = \$ 400,

Time = 3 years.

$$\text{Rate} = (100 \times \text{Interest}) / (\text{Principal} \times \text{Time})$$

$$= (100 \times 400) / (3000 \times 3)$$

$$= 40000/9000$$

$$= 4.44 \%$$

Therefore, Rate = 4.44 %.

2. Find Rate, when Principal = \$ 2100; Interest = \$ 350; Time = 2 years.

Solution:

Principal = \$ 2100,

Interest = \$ 350,

Time = 2 years.

$$\text{Rate} = (100 \times \text{Interest}) / (\text{Principal} \times \text{Time})$$

$$= (100 \times 350) / (2100 \times 2)$$

$$= 35000/4200$$

$$= 8.33 \%$$

Therefore, Rate = 8.33 %.

3. Find Rate, when Principal = \$ 10000; Interest = \$ 800; Time = 5 years.

Solution:

Principal = \$ 10000,

Interest = \$ 800,

Time = 5 years.

$$\text{Rate} = (100 \times \text{Interest}) / (\text{Principal} \times \text{Time})$$

$$= (100 \times 800) / (10000 \times 5)$$

$$= 80000/50000$$

$$= 1.6 \%$$

Therefore, Rate = 1.6 %.

Finding the Number of Years, N

To find Time when Principal Interest and Rate are given the rules are.

$$\text{Interest} = (\text{Principal} \times \text{Rate} \times \text{Time})/100$$

$$\text{Time} = (100 \times \text{Interest})/ (\text{Principal} \times \text{Rate})$$

Examples to find Time (T) when Principal (P), Interest (I) and Rate (R) are given:

1. Find Time when, Principal = \$ 1500; Interest = \$ 450; Rate = 5% p.a.

Solution:

Principal = \$ 1500,

Interest = \$ 450,

Rate = 5% p.a.

$$\text{Time} = (100 \times \text{Interest})/(\text{Principal} \times \text{Rate})$$

$$= (100 \times 450)/(1500 \times 5)$$

$$= 45000/7500$$

$$= 6.$$

Therefore, Time (T) = 6 years.

2. Find Time when, Principal = \$ 2000; Interest = \$ 600; Rate = 3% p.a.

Solution:

Principal = \$ 2000,

Interest = \$ 600,

Rate = 3% p.a.

$$\text{Time} = (100 \times \text{Interest})/(\text{Principal} \times \text{Rate})$$

$$= (100 \times 600)/(2000 \times 3)$$

$$= 60000/6000$$

$$= 10$$

Therefore, Time (T) = 10 years.

3. Find Time when, Principal = \$ 3500; Interest = \$ 700; Rate = 4% p.a.

Solution:

Principal = \$ 3500,

Interest = \$ 700,

Rate = 4% p.a.

Time = $(100 \times \text{Interest}) / (\text{Principal} \times \text{Rate})$

= $(100 \times 700) / (3500 \times 4)$

= 70000/14000

= 5

Therefore, Time (T) = 5 years.

Annuities:

An annuity is a contract between you and an insurance company in which you make a lump-sum payment or series of payments and, in return, receive regular disbursements, beginning either immediately or at some point in the future.

Annuity due VS Ordinary Annuity:

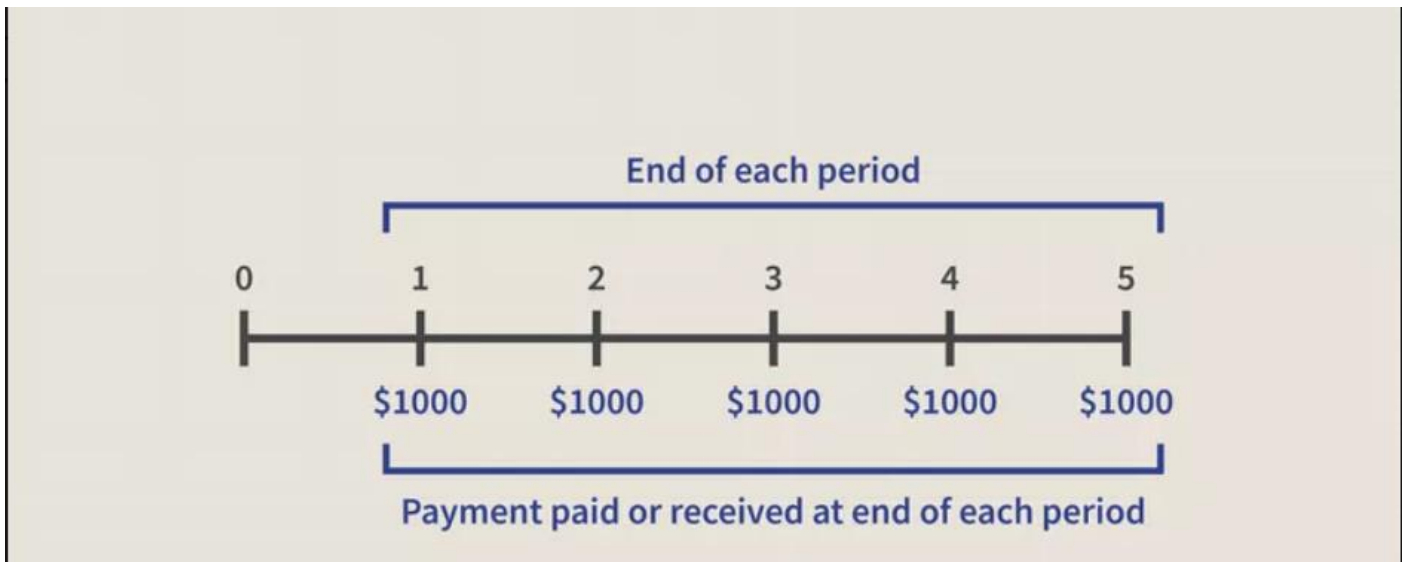
An ordinary annuity is when a payment is made at the end of a period. An annuity due is when a payment is due at the beginning of a period. While the difference may seem meager, it can make a significant impact on your overall savings or debt payments.

- **Ordinary annuities:** An ordinary annuity makes (or requires) payments at the end of each period. For example, bonds generally pay interest at the end of every six months.
- **Annuities due:** With an annuity due, by contrast, payments come at the beginning of each period. Rent, which landlords typically require at the beginning of each month, is a common example.

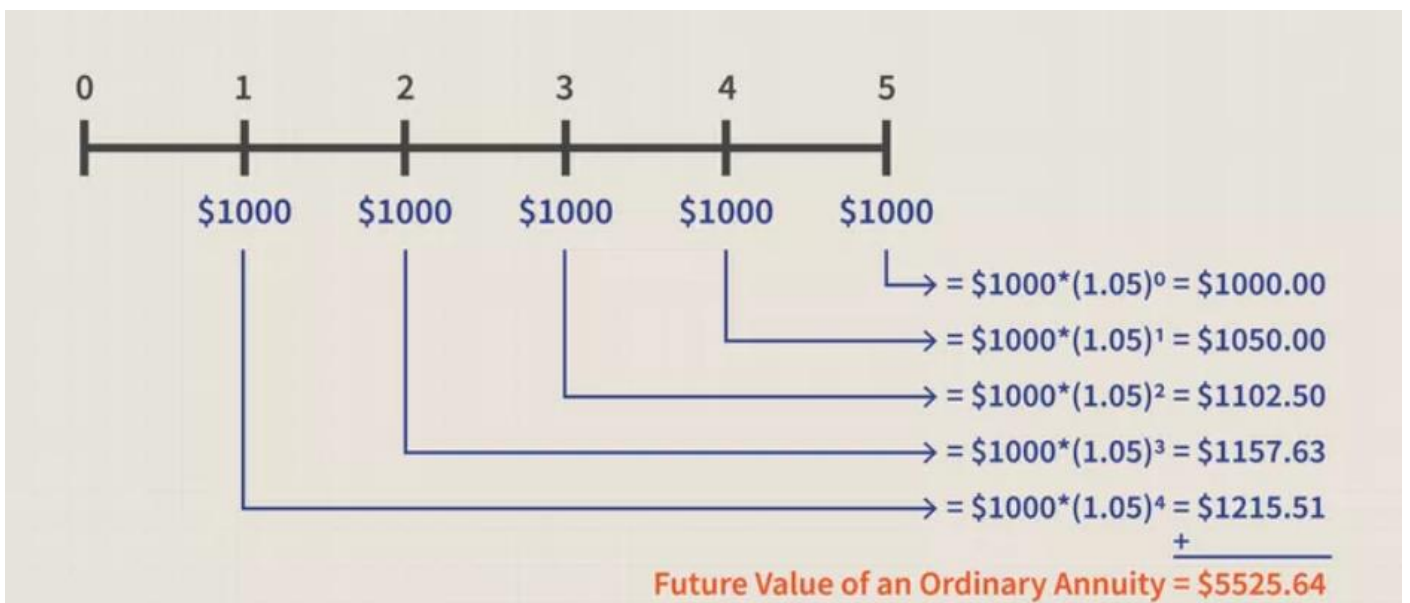
Calculating the Future Value of an Ordinary Annuity:

Consider, for example, a series of five \$1,000 payments made at regular intervals for five years.

Calculate FV of Ordinary Annuity for Five Years.



Because of the time value of money—the concept that any given sum is worth more now than it will be in the future because it can be invested in the meantime—the first \$1,000 payment is worth more than the second, and so on. So, let's assume that you invest \$1,000 every year for the next five years, at 5% interest. Below is how much you would have at the end of the five-year period.



Rather than calculating each payment individually and then adding them all up, however, you can use the following formula, which will tell you how much money you'd have in the end:

$$\text{FV Ordinary Annuity} = C \times (1 + i)^{n-1} / i$$

Where=cash flow per period

i =interest rate

n =number of payments

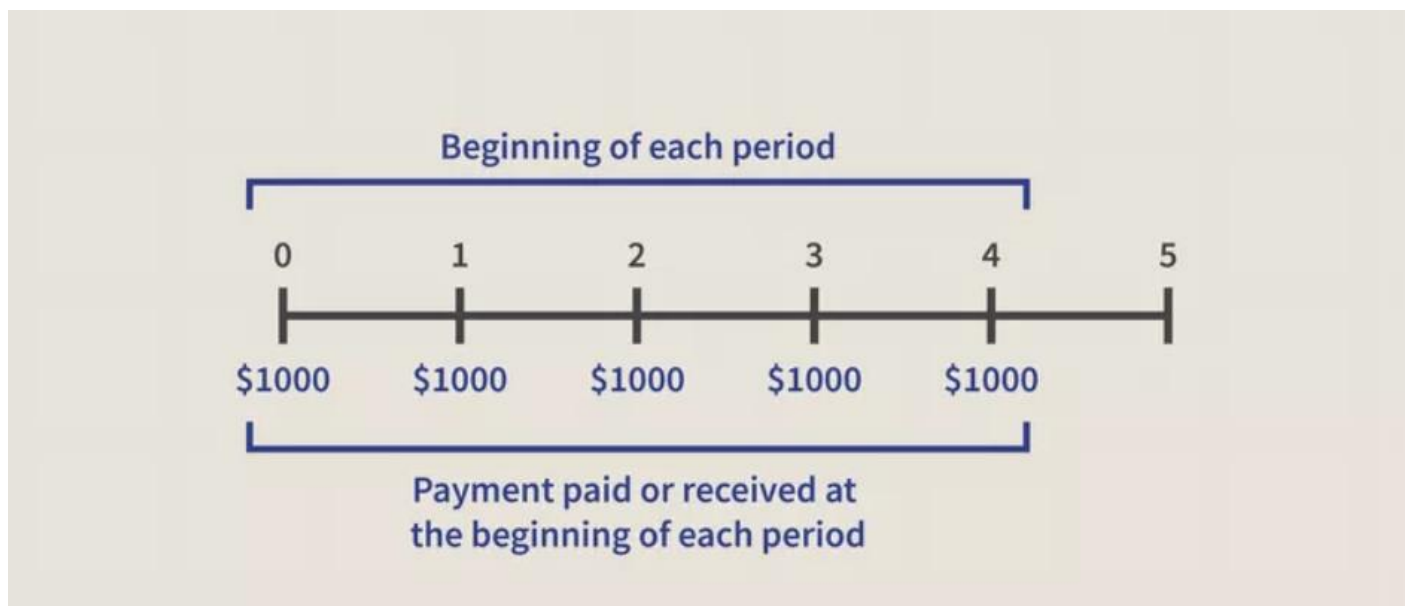
$$\text{FV ordinary Annuity} = \$1,000 \times [(1+0.05)^5 - 1 / 0.05]$$

$$= \$1,000 \times 5.53$$

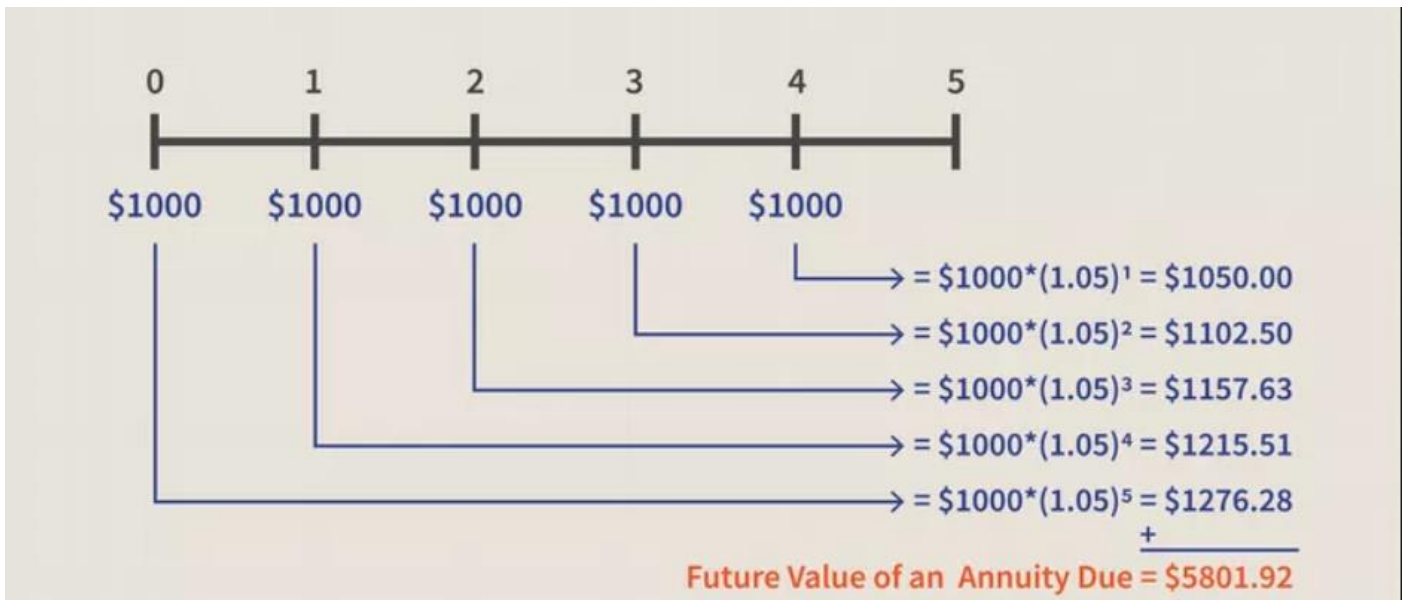
$$= \$5,525.63$$

Calculating the Future Value of an Annuity Due

An annuity due, you may recall, differs from an ordinary annuity in that the annuity due's payments are made at the beginning, rather than the end, of each period.



To account for payments occurring at the beginning of each period, it requires a slight modification to the formula used to calculate the future value of an ordinary annuity and results in higher values, as shown below.



The reason the values are higher is that payments made at the beginning of the period have more time to earn interest. For example, if the \$1,000 was invested on January 1 rather than January 31 it would have an additional month to grow.

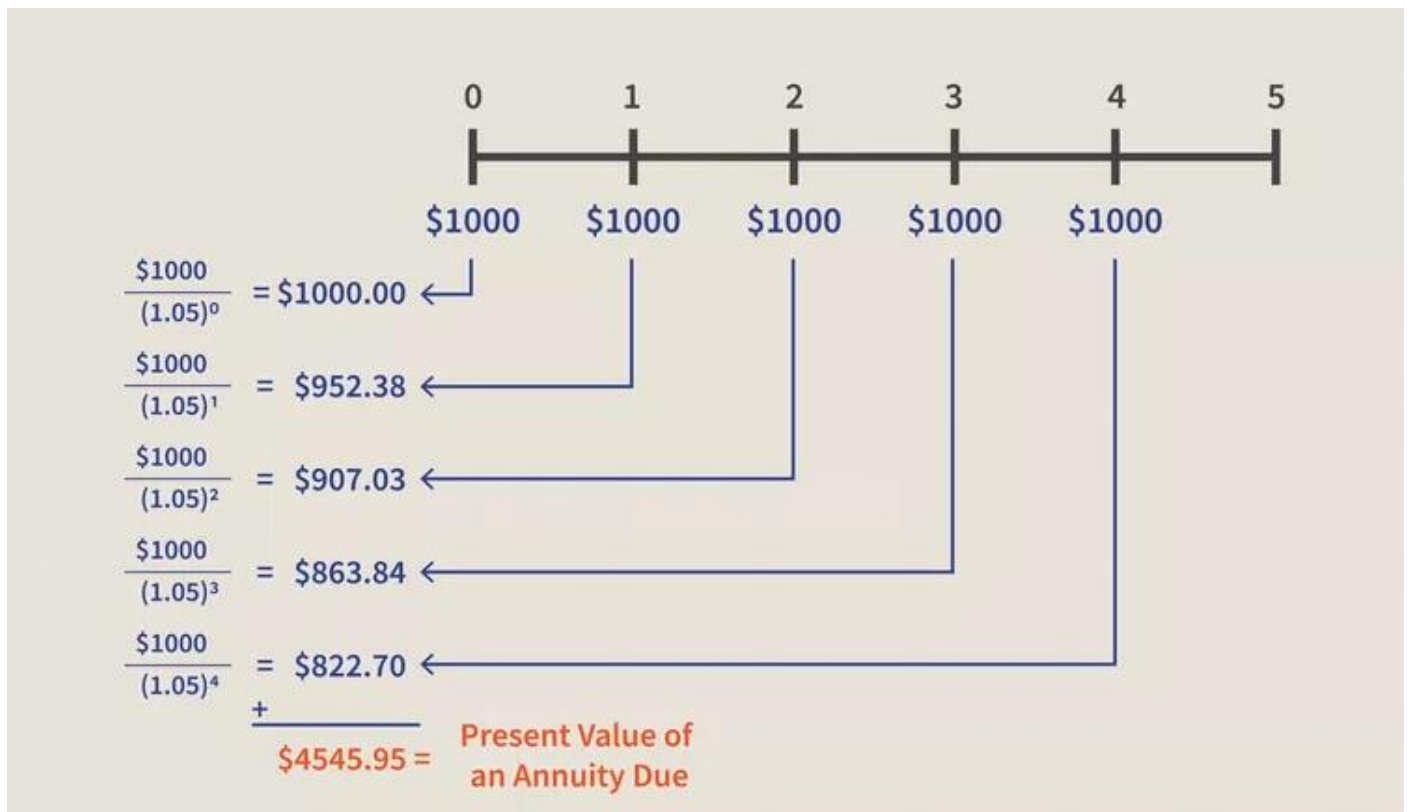
The formula for the future value of an annuity due is as follows:

$$\begin{aligned}
 FV_{\text{Annuity Due}} &= C \times [(1+i)^n - 1/i] \times (1+i) \\
 FV_{\text{Annuity Due}} &= \$1,000 \times [(1+0.05)^5 - 1/0.05] \times (1+0.05) \\
 &= \$1,000 \times 5.53 \times 1.05 \\
 &= \$5,801.91
 \end{aligned}$$

Calculating the Present Value of an Annuity Due:

Similarly, the formula for calculating the present value of an annuity due takes into account the fact that payments are made at the beginning rather than the end of each period.

For example, you could use this formula to calculate the present value of your future rent payments as specified in your lease. Let's say you pay \$1,000 a month in rent. Below, we can see what the next five months would cost you, in terms of present value, assuming you kept your money in an account earning 5% interest.



This is the formula for calculating the present value of an annuity due:

$$\text{PVAnnuity Due} = C \times [1 - (1+i)^{-n} / i] \times (1+i)$$

$$= \$1,000 \times [0.05(1 - (1+0.05)^{-5}) \times (1+0.05)]$$

$$= \$1,000 \times 4.33 \times 1.05$$

$$= \$4,545.95$$

Uneven Cash Flows

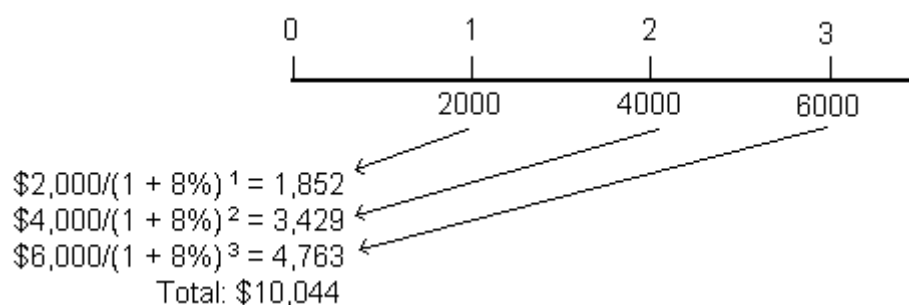
Basically, uneven cash flows refer to a series of unequal payments made over a given period of time. For example, one may receive the following annual payments over a five year period: \$500 US Dollars (USD), \$300 USD, \$400 USD, \$250 USD and \$750 USD. On the other hand, if the regular payments were fixed to a particular amount, then the cash flows would be equal. For instance, one may receive a \$500 USD annual payment, which is also known as an annuity. Moreover, uneven cash flows may be associated with all sorts of financial situations including budgeting.

Future Value of Uneven Cash Flows:

Once we know the present value of the cash flows, we can easily apply time-value equivalence by using the formula to calculate the future value of a single sum of money (LOS a).

Example:

John wants to pay off his student loan in three annual installments: \$2,000, \$4,000 and \$6,000, respectively, in the next three years. How much should John deposit into his bank account today if he wants to use the account balance to pay off the loan? Assume that the bank pays 8% interest, compounded annually.



Semiannual and other Compounding Periods

Compounding interest semiannually means that the principal of a loan or investment at the beginning of the compounding period, in this case, every six months, includes the total interest from each previous period. In simple interest loans and investments, the amount of interest owed is based only on the initial principal amount. When interest is compounded, the interest from every previous period is added to the principal. In compound interest loans, you are paying interest on interest. If you have an investment that compounds interest, you are paid interest on the interest.

Calculate Semiannual Compounded Interest:

$$P [(1+i)^n - 1]$$

Loan Example:

In this example to illustrate how you calculate interest compounded semiannually, you have a loan for \$10,000 with a nominal interest rate of 5%. You will be paying it back over three years. Here is how you determine how much interest you will pay over the life of the loan.

$$P [(1+r)^n - 1] = 10,000[(1+.05)^5 - 1]$$

$$10,000[(1.05)^5 - 1]$$

$$10,000(1.276 - 1)$$

$$10,000(.276)$$

2,760 is the amount of the interest you will pay over the life of the loan.

Add the principal, \$10,000 and the total interest that accrues over the life of your loan, \$2,760. The total amount you will pay over the life of the loan is \$12,760.

Comparing Interest Rates

Simple Interest Rate

If you take out a \$300,000 loan from the bank and the loan agreement stipulates that the interest rate on the loan is 4% simple interest, this means that you will have to pay the bank the original loan amount of \$300,000 + (4% x \$300,000) = \$300,000 + \$12,000 = \$312,000.

The example above was calculated based on the annual simple interest formula, which is:

Simple interest = **principal X interest rate X time**

The individual that took out a loan will have to pay \$12,000 in interest at the end of the year, assuming it was only a one-year lending agreement. If the term of the loan was a 30-year mortgage, the interest payment will be:

$$\text{Simple interest} = \$300,000 \times 4\% \times 30 = \$360,000$$

A simple interest rate of 4% annually translates into an annual interest payment of \$12,000. After 30 years, the borrower would have made \$12,000 x 30 years = \$360,000 in interest payments, which explains how banks make their money.

Prepared By: Waqar Ali

Lecturer Riphah International College

Compound Interest Rate

Some lenders prefer the compound interest method, which means that the borrower pays even more in interest. Compound interest, also called interest on interest, is applied both to the principal and also to the accumulated interest made during previous periods. The bank assumes that at the end of the first year the borrower owes the principal plus interest for that year. The bank also assumes that at the end of the second year, the borrower owes the principal plus the interest for the first year plus the interest on interest for the first year.

The interest owed when compounding is higher than the interest owed using the simple interest method. The interest is charged monthly on the principal including accrued interest from the previous months. For shorter time frames, the calculation of interest will be similar for both methods. As the lending time increases, however, the disparity between the two types of interest calculations grows.

Using the example above, at the end of 30 years, the total owed in interest is almost \$700,000 on a \$300,000 loan with a 4% interest rate.

The following formula can be used to calculate compound interest:

$$\text{Compound interest} = P [(1 + \text{interest rate})^n - 1]$$

where:

p = principal

n = number of compounding periods